

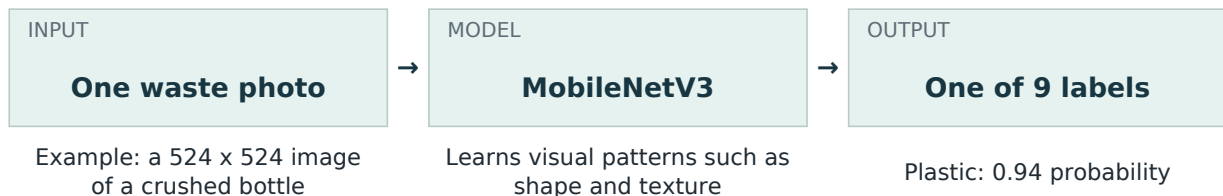
RealWaste: Image-Based Waste Material Classification

Machine Learning Project Proposal

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1. Problem and planned system

Waste sorting is difficult when objects are dirty, crushed, mixed with other materials, or photographed in a busy landfill. Incorrect sorting can contaminate recycling streams and send reusable material to landfill. We propose a model that recognizes the main material in a real waste photograph.



Task type and target

This is multiclass image classification. The input is a color image. The target is its folder label: Cardboard, Food Organics, Glass, Metal, Miscellaneous Trash, Paper, Plastic, Textile Trash, or Vegetation. The output is nine probabilities and the label with the highest probability.

What the final demonstration will look like

- The user selects a waste image.
- The model resizes it to 224 x 224 pixels and analyzes it.
- The screen shows the predicted material and probability, such as Metal: 82%.

A prediction should support a human sorter. It should not automatically decide hazardous-waste handling because the dataset has no hazard labels.

2. Dataset summary

IMAGES	CLASSES	IMAGE SIZE
4,752	9	524 x 524

RealWaste was collected at the Whyte's Gully Waste and Resource Recovery facility in Wollongong, Australia. It contains color photographs taken in an authentic landfill setting. UCI reports no missing images. The compressed download is approximately 656.6 MB.

Source: [UCI RealWaste dataset page](#); DOI: [10.24432/C5SS4G](#).

Dataset inspection and required visuals

Actual samples inspected

UCI lists the individual files, dimensions, and sizes. The entries below are real files shown on the dataset page. They confirm the folder-based label and the 524 x 524 image format.

Actual file entries inspected on the UCI dataset page

524 x 524 color image Vegetation_109.jpg 242.8 KB	524 x 524 color image Vegetation_110.jpg 242.7 KB	524 x 524 color image Vegetation_111.jpg 241.0 KB
524 x 524 color image Vegetation_107.jpg 239.6 KB	524 x 524 color image Vegetation_108.jpg 239.5 KB	

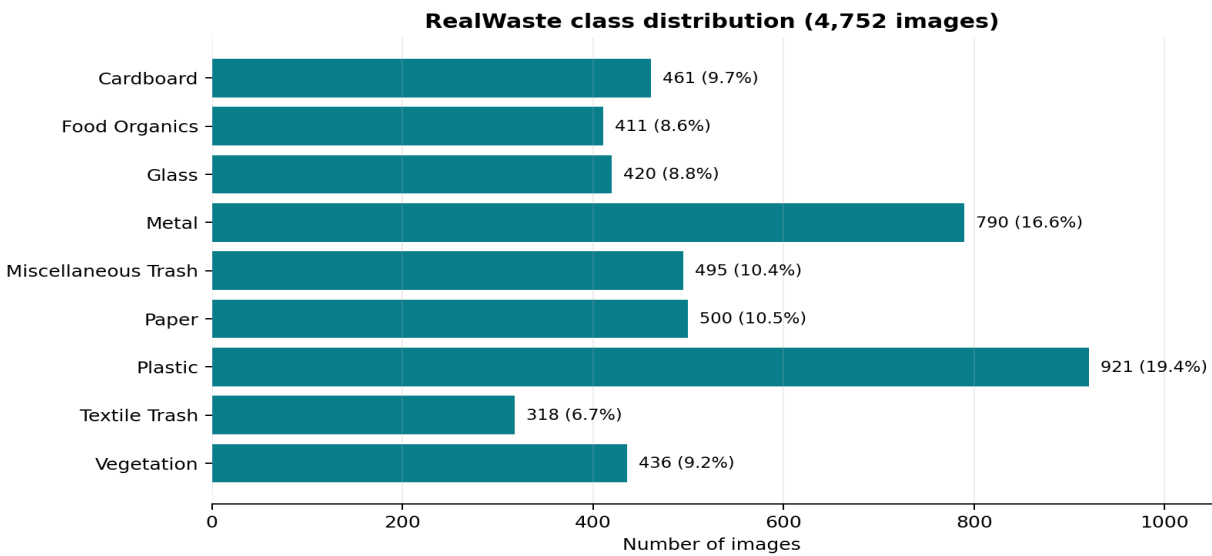
Image pixels are not reproduced here. analysis.py creates an actual one-per-class contact sheet after download.

Representative class visualization

The repository uses one folder for each material. The included **analysis.py** selects one real image from every folder and creates a 3 x 3 contact sheet. This avoids copying the 656.6 MB dataset into the submission archive while keeping the inspection reproducible.

Cardboard	Food Organics	Glass
Metal	Misc. Trash	Paper
Plastic	Textile Trash	Vegetation

Class distribution



Plastic is the largest class with 921 images (19.4%). Textile Trash is the smallest with 318 (6.7%). The imbalance is moderate, so the metric should not allow large classes to hide poor results on small classes.

Evaluation plan

3. Train, validation, and test split

Before training, we will use a fixed random seed and a stratified 70% / 15% / 15% split. Stratified means every split keeps roughly the same percentage of each material. The table shows the planned approximate allocation; the code records the exact file list in **split_manifest.csv**.

Class	Total	Train ~70%	Validation ~15%	Test ~15%
Cardboard	461	323	69	69
Food Organics	411	288	62	61
Glass	420	294	63	63
Metal	790	553	119	118
Miscellaneous Trash	495	347	74	74
Paper	500	350	75	75
Plastic	921	645	138	138
Textile Trash	318	223	48	47
Vegetation	436	305	65	66
Approx. total	4,752	3,328	713	711

Leakage controls

- Search for exact and near-duplicate images before splitting. Similar copies must stay in the same split.
- Apply cropping, flipping, and color changes only to training images.
- Use validation data to choose settings. Open the test set only for the final result.

4. Single validation metric

Macro F1

A score from 0 to 1 that calculates performance for each class separately, then gives all nine classes equal importance. Higher is better.

Macro F1 is better than accuracy here because a model could perform well on Plastic and Metal while failing on Textile Trash. Only validation macro F1 will be used to compare models.

Model and baseline

We will first use MobileNetV3 Small with pretrained image knowledge, replace its final layer with nine outputs, and fine-tune it on the training images. A simple majority-class baseline always predicts Plastic. It would be correct only 19.4% of the time and gives a weak macro F1, so the trained model must clearly beat it.

Access, license, and limitations

UCI lists CC BY 4.0, but the creators' GitHub repository states CC BY-NC-SA 4.0. We will follow the stricter CC BY-NC-SA 4.0 terms: academic, noncommercial use; attribution; and share-alike conditions if data derivatives are redistributed. The model may learn landfill backgrounds as well as materials, and results may not transfer to household bins without new testing.

Sources

1. UCI: archive.ics.uci.edu/dataset/908/realwaste
2. Creators' repository: github.com/sam-single/realwaste
3. Single, Iranmanesh, and Raad (2023): doi.org/10.3390/info14120633